

Inflation Persistence After an Energy-Price Shock: New Keynesian Model with Vertical Production Networks and Real Wage Rigidity

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EC424 Extended Essay

Observations in the UK

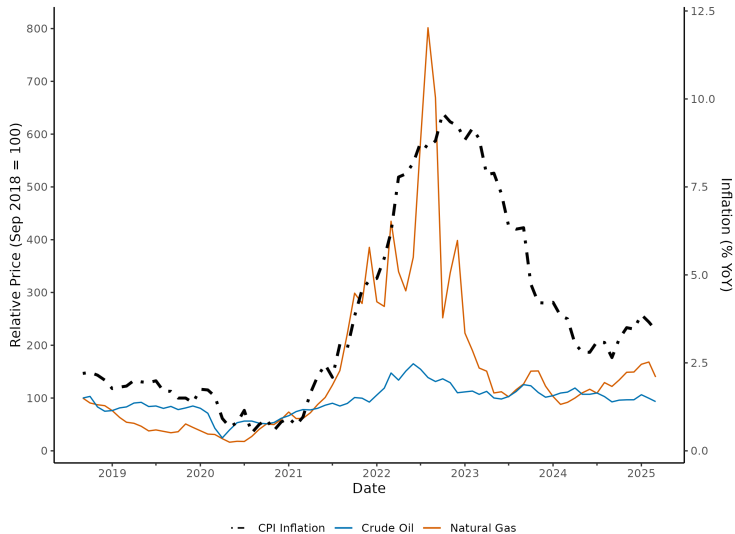


Figure 1: Energy Prices and UK CPI Inflation (Sep 2018 - Mar 2025)

What happened in the UK after the gas price spike?

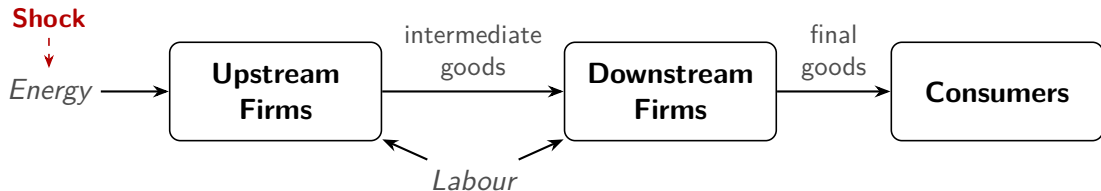
- ▶ Natural gas price decreased relatively quickly; CPI inflation decreased **more slowly**.
- ▶ **Few months lag** between the peak of gas price and the peak of CPI inflation.
- ▶ CPI inflation did not return to the pre-shock level¹

¹I do not feature this in my model in fear of simultaneous trends.

Three Model Extensions from the Baseline

1. Production network

- ▶ Structured vertically with upstream and downstream industries
 - ▶ Upstream industry: energy + labour $\rightarrow$ intermediate goods
 - ▶ Downstream industry: intermediate goods + labour $\rightarrow$ final consumption goods
- ▶ Production function remains Leontief for both sectors
- ▶ Competitive sector remains unchanged, producing tradable goods using labour only for the balance of payments equilibrium
- ▶ Single labour market with free mobility ($\Rightarrow$ equal wage across sectors)



2. **Myopia** in downstream firms' Calvo reset price determination

- ▶ Minton and Wheaton (2023); Gabaix (2020)
- ▶ Firms are reluctant / difficult to forecast the future, and thus discount the future expectations
- ▶ The myopic expectation of $\hat{v}_{t+k}$ (in log-linearised form) is

$$\tilde{\mathbb{E}}_t[\hat{v}_{t+k}] \equiv m_f^k \mathbb{E}_t[\hat{v}_{t+k}],$$

- ▶ where $m_f \in [0, 1]$ with $m_f = 1$ being completely rational

3. **Real wage rigidity** in the labour market

- ▶ Parsimonious way to represent labour market frictions, such as hiring costs, à la Blanchard and Galí (2007)
- ▶ Only a fraction $1 - \psi_w$ of the gap between the actual real wage $\hat{w}_t$ and the flexible real wage $\hat{w}_t^f$ is adjusted in each period:

$$\hat{w}_t = \psi_w \hat{w}_{t-1} + (1 - \psi_w) \hat{w}_t^f,$$

- ▶ where $\psi_w \in [0, 1]$ with $\psi_w = 0$ being completely flexible

Log-Linearised Model Equations 1/2

$$\hat{y}_t = \hat{L}_t - \alpha^U \alpha^D \hat{s}_t \quad (\text{aggregate production function})$$

$$\hat{x}_t^D = \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1 - \Phi_{DU})\hat{w}_t \quad (\text{downstream real marginal cost})$$

$$\hat{X}_t^D = \hat{x}_t^D + \hat{p}_t^D \quad (\text{downstream nominal marginal cost})$$

$$\hat{X}_t^U = \alpha^U \hat{s}_t + \hat{w}_t + \hat{p}_t^D \quad (\text{upstream nominal marginal cost})$$

$$\hat{w}_t^f = \frac{1}{\sigma} \hat{y}_t + \frac{1}{\psi} \hat{L}_t \quad (\text{flexible real wage})$$

$$\hat{w}_t = \psi_w \hat{w}_{t-1} + (1 - \psi_w) \hat{w}_t^f \quad (\text{real wage rigidity})$$

Log-Linearised Model Equations 2/2

$$\hat{\pi}_t = \hat{p}_t^D - \hat{p}_{t-1}^D \quad (\text{CPI inflation})$$

$$\hat{p}_t^D = \theta^D \hat{p}_{t-1}^D + (1 - \theta^D) \hat{p}_t^{D*} \quad (\text{downstream Calvo price level})$$

$$\hat{p}_t^{D*} = \beta m_f \theta^D \mathbb{E}_t[\hat{p}_{t+1}^{D*}] + (1 - \beta m_f \theta^D) \hat{X}_t^D \quad (\text{downstream reset price})$$

$$\hat{p}_t^U = \theta^U \hat{p}_{t-1}^U + (1 - \theta^U) \hat{p}_t^{U*} \quad (\text{upstream Calvo price level})$$

$$\hat{p}_t^{U*} = \beta \theta^U \mathbb{E}_t[\hat{p}_{t+1}^{U*}] + (1 - \beta \theta^U) \hat{X}_t^U \quad (\text{upstream reset price})$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]) \quad (\text{IS equation})$$

$$\hat{i}_t = \phi \hat{i}_{t-1} + (1 - \phi)(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t) \quad (\text{Taylor rule})$$

$$\hat{s}_t = \rho \hat{s}_{t-1} + u_t \quad (\text{Terms of trade AR(1) process})$$

Parameter Calibration

Parameter	Symbol	Value
<i>Households</i>		
Discount factor	β	0.9984
Elasticity of intertemporal substitution	σ	0.5
Frisch elasticity of labour supply	ψ	3
<i>Production and cost structure</i>		
Energy cost share (upstream)	α^U	0.375
Intermediate input share (downstream)	α^D	0.4
Elasticity of substitution	ε	11
Downstream's intermediate good-cost share	Φ_{DU}	0.4231 (implied)
<i>Price setting</i>		
Downstream Calvo probability	θ^D	11/12
Upstream Calvo probability	θ^U	3/4
Myopia parameter (downstream)	m_f	0.5
<i>Labour market</i>		
Real wage rigidity	ψ_w	$0.7^{1/3} \approx 0.888$
<i>Monetary policy</i>		
Interest-rate smoothing	ϕ	0.9
Inflation response	χ_π	3
Output response	χ_y	0.5
<i>Shock process</i>		
Terms of trade persistence	ρ	0.8

Energy cost share (upstream): $\alpha^U = 0.375$

- ▶ Output-weighted ave. energy cost share across eight most energy-intensive sectors²
 - ▶ Model's weak point: those sectors produce final consumption goods in reality
- ▶ Based on the input-output table of the UK economy 2017-2022 by ONS

Intermediate input share (downstream): $\alpha^D = 0.4$

- ▶ So that the energy cost share in the final consumption goods is $\alpha^U \alpha^D = 0.15$

Downstream's intermediate good-cost share: $\Phi_{DU} = 0.4231$ (implied)

$$\Phi_{DU} = \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu} = 0.4231$$

- ▶ Cost-based share of intermediate goods in the downstream real marginal cost
- ▶ Takes into account the upstream industry's markup on the intermediate goods
 - ▶ cf. $\alpha^D = 0.4$ does not

²Electric Power Generation, Gas Manufacture & Distribution, Industrial Gases, Inorganics, Fertilisers, Petrochemicals, Paper and Paper Products, Glass, Ceramics, Stone, Basic Iron and Steel, and Other Basic Metals & Casting

Upstream Calvo probability: $\theta^U = 3/4$

- ▶ Average price duration of 4 months
- ▶ Nakamura and Steinsson (2008): prices for intermediate goods are more flexible than those for final consumption goods
- ▶ Sensitivity analysis with $\theta^U \in \{0.5, 3/4, 11/12\}$

Myopia parameter (downstream): $m_f = 0.5$

- ▶ The mid-point of the parameter space $[0, 1]$ given the variety of intermediate goods
- ▶ Sensitivity analysis with $m_f \in \{0, 0.5, 1\}$

Real wage rigidity: $\psi_w = 0.7^{1/3} \approx 0.888$

- ▶ The mid point of the two sample values in Blanchard and Galí (2007) in monthly frequency
- ▶ Sensitivity analysis with $\psi_w \in \{0, 0.7^{1/3}, 0.9^{1/3}\}$

Other parameters are calibrated to standard values in the literature.

Results

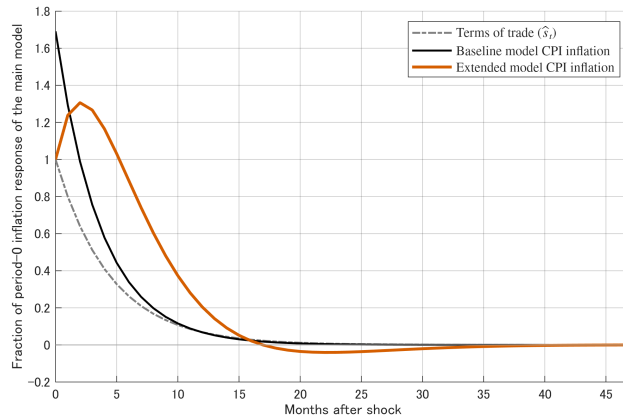


Figure 2: Inflation Persistence vs Energy Price Shock Decay

- CPI inflation of both models is normalised based on my model's initial response

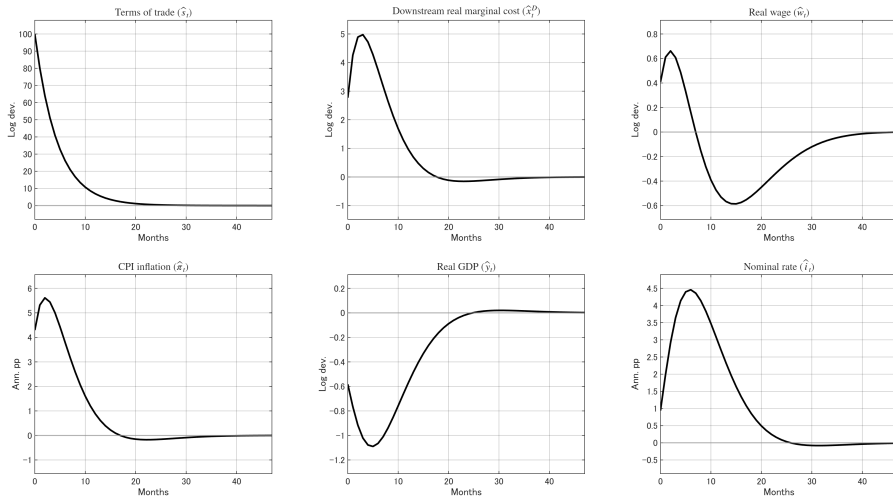


Figure 3: Impulse Responses of the Model to an Exogenous Terms of Trade Shock

Downstream Real Marginal Cost $\hat{x}_t^D$

- ▶ peaks at 5.0% with a three-month lag
- ▶ Lag is due to the upstream Calvo pricing + real wage rigidity

Real Wage $\hat{w}_t$

- ▶ High $\hat{s}_t$ demands more labour for the competitive sector
 - ▶ This offsets the negative effect due to the productivity decrease
- ▶ After the peak, nominal interest rate will be raised to stabilise inflation, which decreases labour demand
 - ▶ But adjustment is slow due to real wage rigidity

CPI Inflation $\hat{\pi}_t$

- ▶ Peaks at 5.6% with a three-month lag
 - ▶ Myopia of the downstream firms contributes to the lag and persistence

Sensitivity Analysis (IRFs in Appendix)

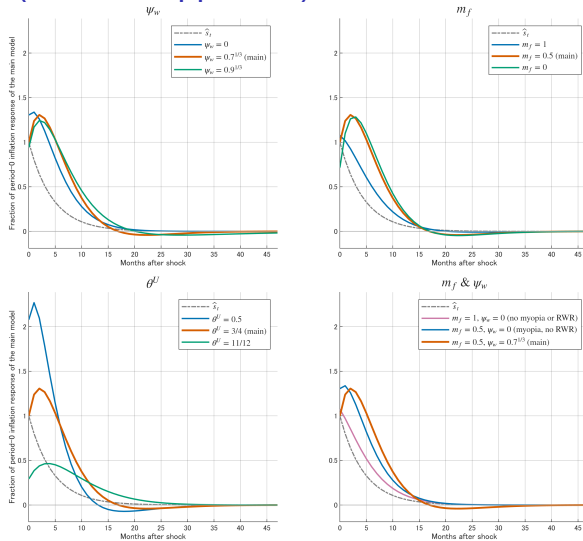


Figure 4: Sensitivity Analysis Results

Real Wage Rigidity ψ_w

- ▶ No real wage rigidity ($\psi_w = 0$):
 - ▶ Inflation responds immediately with initial response almost as high as its peak in one month
- ▶ Higher real wage rigidity ($\psi_w = 0.9^{1/3}$):
 - ▶ Slightly smaller response peak
 - ▶ But inflation is more persistent with a slower decay

Downstream Firm Myopia m_f

- ▶ No myopia / completely rational ($m_f = 1$):
 - ▶ Inflation decreases monotonically after the initial shock
 - ▶ Correct expectations that energy price will only decrease back to the steady state level
- ▶ Full myopia ($m_f = 0$):
 - ▶ Initial inflation response is smaller
 - ▶ but catches up in four months to reach the almost same peak

Upstream Calvo Probability θ^U

- ▶ $\theta^U = 0.5$ (average price duration of 2 months):
 - ▶ Twice as large initial response as the original case
 - ▶ Inflation decreases much faster afterwards
 - ▶ Upstream firms are more able to pass through the energy price shock
 - ▶ Vice versa for less flexible prices with $\theta^U = 11/12$

Combination: Myopia m_f and Real Wage Rigidity ψ_w

- ▶ No myopia and no real wage rigidity ($m_f = 0, \psi_w = 0$):
 - ▶ Vertical production network with Calvo pricing for both industries
 - ▶ No lag in the response peak nor inflation persistence
- ▶ Myopia but no real wage rigidity ($m_f = 0.5, \psi_w = 0$):
 - ▶ Higher and more persistent inflation but the peak lag of only one month
 - ▶ Myopia alone cannot generate enough long lag observed in the data
- ▶ Myopia and real wage rigidity ($m_f = 0.5, \psi_w = 0.7^{1/3}$, the original case):
 - ▶ 3-month lag in the response peak, after which inflation is more persistent
 - ▶ The peak is only as high as the previous case with no real wage rigidity, but importantly the response is hump-shaped

Discussions and Conclusion

Three extensions combined can generate a lag in the inflation response peak & persistence as observed in the UK

- ▶ Vertical production network with Calvo pricing for both industries
- ▶ Myopia in downstream firms' price setting
- ▶ Real wage rigidity

Limitations

- ▶ Vertical structure of the production network
 - ▶ In reality, what I have classified as the “upstream industry” also produces final consumption goods
- ▶ Inflation goes back to zero in the model, while the UK inflation remains around 3%
 - ▶ There may be simultaneous trends in reality, though
- ▶ I haven't discussed about the absolute magnitude of the shock / response
 - ▶ Difficult to identify the size of the energy price shock in practice?
- ▶ Choice of and reasoning behind the parameter values
 - ▶ Particularly, $\theta^U = 3/4$, $\theta^D = 11/12$, $m_f = 0.5$, $\psi_w = 0.7^{1/3}$
 - ▶ I believe those are “theoretically” somewhat reasonable, but not necessarily supported by empirical micro evidence (Zhou and Dixon 2019; Bunn and Ellis 2012a, 2012b)

Practical questions

- ▶ How much should I describe the baseline model?
- ▶ Should I derive the log-linearisation in the appendix?
- ▶ Do parameter value choices make sense? Is more reasoning required?
 - ▶ Picking a single country doesn't mean much? As many parameters are not directly measured in data?

References

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Appendix

The Baseline Model

$$\hat{y}_t = \hat{L}_t - \alpha \hat{s}_t \quad (\text{aggregate production function})$$

$$\hat{x}_t = \hat{w}_t + \alpha \hat{s}_t \quad (\text{real marginal cost})$$

$$\hat{X}_t = \hat{x}_t + \hat{P}_t \quad (\text{nominal marginal cost})$$

$$\hat{w}_t = \frac{1}{\sigma} \hat{y}_t + \frac{1}{\psi} \hat{L}_t \quad (\text{labour supply})$$

$$\hat{\pi}_t = \hat{P}_t - \hat{P}_{t-1} \quad (\text{inflation})$$

$$\hat{P}_t = \theta \hat{P}_{t-1} + (1 - \theta) \hat{P}_t^* \quad (\text{Calvo price level})$$

$$\hat{P}_t^* = \beta \theta \mathbb{E}_t[\hat{P}_{t+1}^*] + (1 - \beta \theta) \hat{X}_t \quad (\text{reset price})$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]) \quad (\text{IS equation})$$

$$\hat{i}_t = \phi \hat{i}_{t-1} + (1 - \phi)(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t) \quad (\text{Taylor rule})$$

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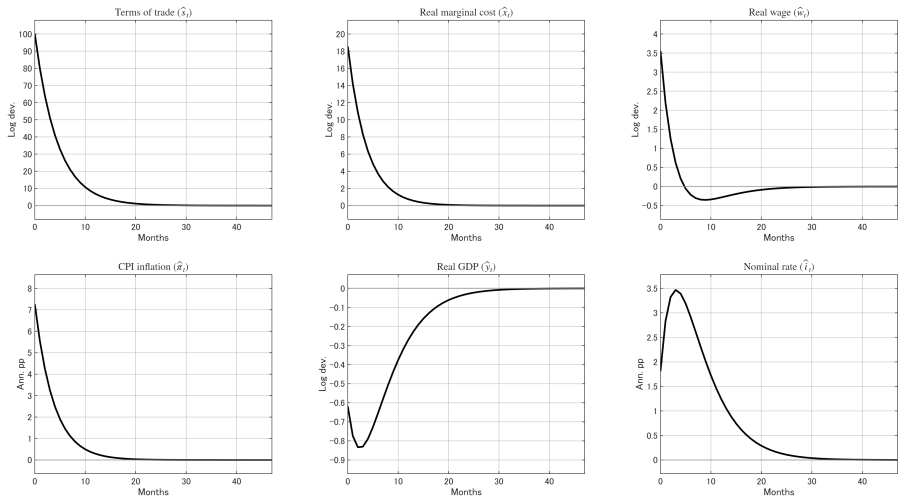


Figure 5: Impulse Responses of the Baseline Model to an Exogenous Terms of Trade Shock

Impulse Responses from Sensitivity Analysis

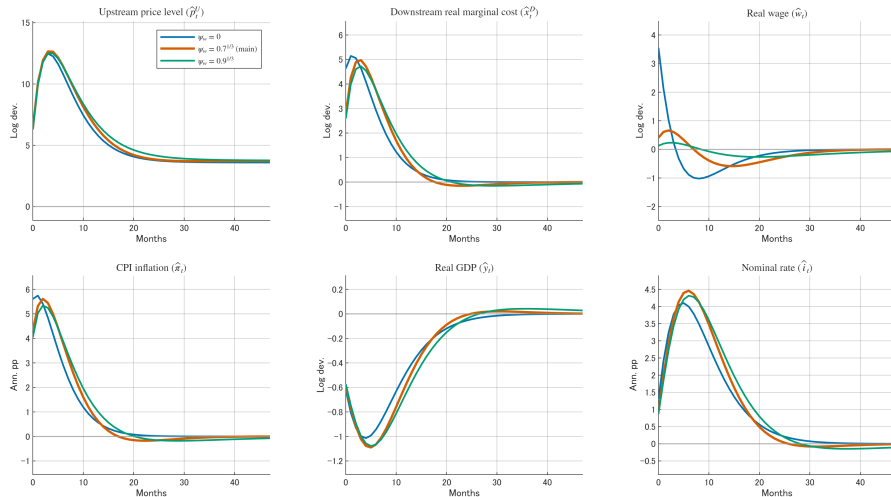


Figure 6: Impulse Responses for Varying Real Wage Rigidity ψ_w

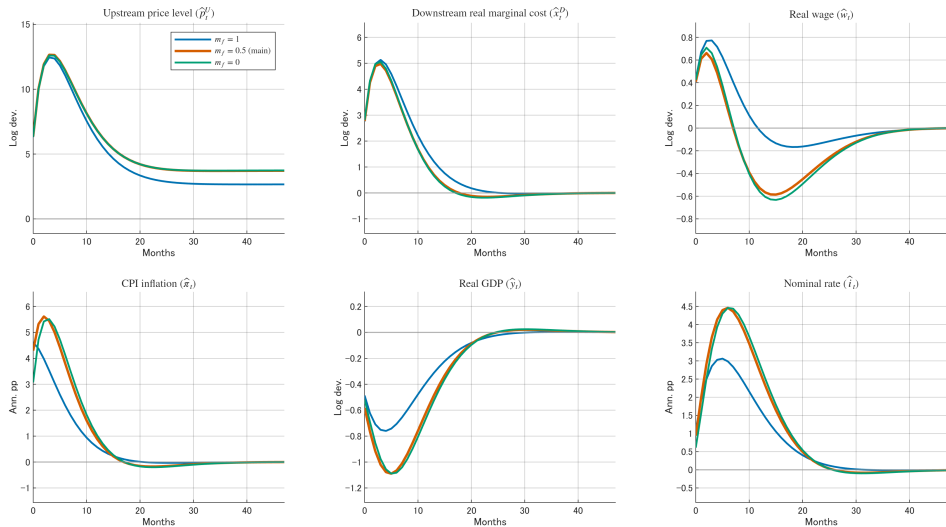


Figure 7: Impulse Responses for Varying Myopia m_f

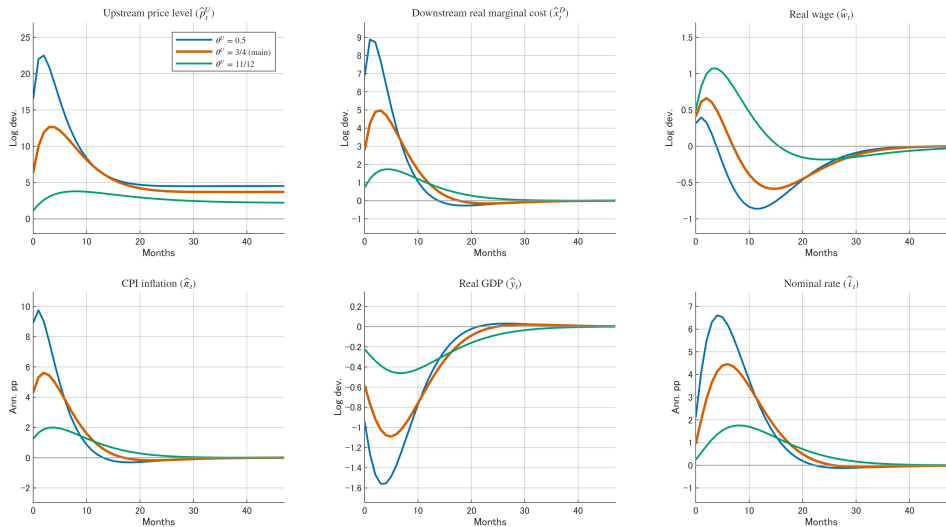


Figure 8: Impulse Responses for Varying Upstream Calvo Probability θ^U

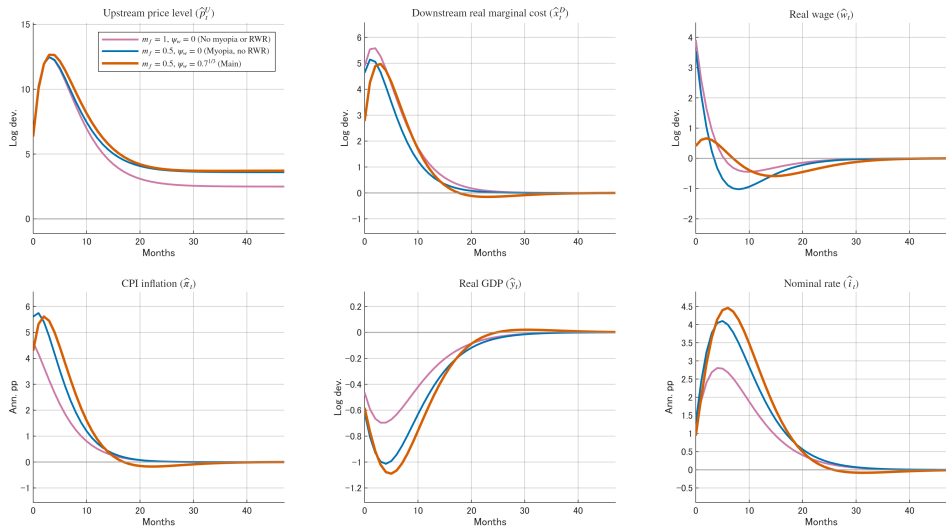


Figure 9: Impulse Responses for Different Combinations of m_f and ψ_w